

Generalization

Out-of-domain robustness, in-the-wild evaluation and corruption

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1 Does Audio Deepfake Detection Generalize?

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Interspeech 2022 · arXiv:2203.16263

■ <https://arxiv.org/abs/2203.16263>

ABSTRACT

Current text-to-speech algorithms produce realistic fakes of human voices, making deepfake detection a much-needed area of research. While researchers have presented various techniques for detecting audio spoofs, it is often unclear exactly why these architectures are successful: Preprocessing steps, hyperparameter settings, and the degree of fine-tuning are not consistent across related work. Which factors contribute to success, and which are accidental? In this work, we address this problem: We systematize audio spoofing detection by re-implementing and uniformly evaluating architectures from related work. We identify overarching features for successful audio deepfake detection, such as using cqt-spec or logspec features instead of melspec features, which improves performance by 37% EER on average, all other factors constant. Additionally, we evaluate generalization capabilities: We collect and publish a new dataset consisting of 37.9 hours of found audio recordings of celebrities and politicians. We find that related work performs poorly on such real-world data (performance degradation of up to one thousand percent). This may suggest that the community has tailored its solutions too closely to the prevailing ASVspoof benchmark and that deepfakes are much harder to detect outside the lab than previously thought.

2 The Codecfake Dataset and Countermeasures for the Universally Detection of Deepfake Audio

Yuantun Xie, Yi Lu, Ruibo Fu, Zhengqi Wen, Zhiyong Wang, Jianhua Tao, Xin Qi, Xiaopeng Wang, Yukun Liu, Haonan Cheng, Long Ye, Yi Sun

IEEE Trans. Audio, Speech & Language Processing 2025 · arXiv:2405.04880

■ <https://arxiv.org/abs/2405.04880>

ABSTRACT

With the proliferation of Audio Language Model (ALM) based deepfake audio, there is an urgent need for generalized detection methods. ALM-based deepfake audio currently exhibits widespread, high deception, and type versatility, posing a significant challenge to current audio deepfake detection (ADD) models trained solely on vocoded data. To effectively detect ALM-based deepfake audio, we focus on the mechanism of the ALM-based audio generation method, the conversion from neural codec to waveform. We initially constructed the Codecfake dataset, an open-source, large-scale collection comprising over 1 million audio samples in both English and Chinese, focus on ALM-based audio detection. As countermeasure, to achieve universal detection of deepfake audio and tackle domain ascent bias issue of original sharpness aware minimization (SAM), we propose the CSAM

strategy to learn a domain balanced and generalized minima. In our experiments, we first demonstrate that ADD model training with the Codefake dataset can effectively detect ALM-based audio. Furthermore, our proposed generalization countermeasure yields the lowest average equal error rate (EER) of 0.616% across all test conditions compared to baseline models. The dataset and associated code are available online.

3 Deepfake-Eval-2024: A Multi-Modal In-the-Wild Benchmark of Deepfakes Circulated in 2024

Nuria Alina Chandra et al.

arXiv:2503.02857 (2025)

■ <https://arxiv.org/abs/2503.02857>

ABSTRACT

While many deepfake detectors report high accuracy on academic datasets, we show that these academic benchmarks are out of date and not representative of real-world deepfakes. We introduce Deepfake-Eval-2024, a new deepfake detection benchmark consisting of in-the-wild deepfakes collected from social media and deepfake detection platform users in 2024. Deepfake-Eval-2024 consists of 45 hours of videos, 56.5 hours of audio, and 1,975 images, encompassing the latest manipulation technologies. The benchmark contains diverse media content from 88 different websites in 52 different languages. We find that the performance of open-source state-of-the-art deepfake detection models drops precipitously when evaluated on Deepfake-Eval-2024, with AUC decreasing by 50% for video, 48% for audio, and 45% for image models compared to previous benchmarks. [Abstract excerpt; see the arXiv listing for the complete text.]

4 Measuring the Robustness of Audio Deepfake Detectors

Xiang Li, Pin-Yu Chen, Wenqi Wei

arXiv:2503.17577 (2025)

■ <https://arxiv.org/abs/2503.17577>

ABSTRACT

Deepfakes have become a universal and rapidly intensifying concern of generative AI across various media types such as images, audio, and videos. Among these, audio deepfakes have been of particular concern due to the ease of high-quality voice synthesis and distribution via platforms such as social media and robocalls. Consequently, detecting audio deepfakes plays a critical role in combating the growing misuse of AI-synthesized speech. However, real-world scenarios often introduce various audio corruptions, such as noise, modification, and compression, that may significantly impact detection performance. This work systematically evaluates the robustness of 10 audio deepfake detection models against 16 common corruptions, categorized into noise perturbation, audio modification, and compression. Using both traditional deep learning models and state-of-the-art foundation models, we make four unique observations.

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